

ULLADULLA, Australia

WMO: 949380

Lat: 35.3636S

Lon: 150.4828E

Elev: 37

StdP: 100.88

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	7.8	-3.0	3.0	13.1	-1.6	3.3	12.9	12.4	12.9	10.5	13.1	3.0	260	0.424

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.6	28.3	18.4	25.9	18.9	24.6	19.6	22.3	24.4	21.7	23.8	21.2	23.3	4.9	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	16.4	23.3	21.0	15.8	23.0	20.4	15.2	22.6	66.0	24.5	63.8	23.9	61.7	23.3	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	9.5	8.2	5.2	35.1	0.9	3.1	4.5	37.3	3.9	39.2	3.4	40.9	2.7	43.2		
			2.3	23.4	0.8	1.0	1.7	24.1	1.2	24.7	0.8	25.3	0.2	26.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	20.4	20.6	19.7	17.5	15.3	13.3	12.5	13.1	14.7	16.2	17.4	19.1
	DBStd	3.54	2.08	1.93	1.96	2.10	2.03	1.73	1.81	2.09	2.60	2.84	2.55	2.26
	HDD10.0	3	0	0	0	0	0	1	2	1	0	0	0	0
	HDD18.3	903	4	3	9	39	97	152	181	162	114	79	46	17
	CDD10.0	2422	323	296	300	225	164	98	78	97	141	193	223	282
	CDD18.3	281	69	66	51	14	3	0	0	1	5	14	19	41
	CDH23.3	718	156	131	86	33	4	0	1	2	31	102	72	100
	CDH26.7	168	37	22	14	7	0	0	0	0	5	36	23	24
Wind	WSAvg	3.7	3.9	3.8	3.8	3.6	3.4	3.6	3.5	3.7	3.8	3.8	3.9	3.8
Precipitation	PrecAvg	1254	95	139	132	120	110	121	61	71	82	106	115	99
	PrecMax	2148	250	649	431	496	360	674	196	394	267	396	500	328
	PrecMin	571	7	10	4	6	0	0	0	0	0	1	1	6
	PrecStd	410	60	132	109	114	85	126	52	86	61	86	90	75
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.2	29.5	28.3	26.8	23.9	20.1	20.5	23.0	27.0	31.4	30.1	30.0
		MCWB	19.2	20.7	19.5	16.3	14.9	13.7	11.8	12.6	14.6	16.8	16.9	18.6
	2%	DB	26.0	25.7	25.1	23.7	21.3	18.5	18.2	19.9	23.2	26.1	24.8	25.8
		MCWB	20.6	21.5	20.1	17.3	15.3	13.3	11.6	12.2	13.8	15.3	17.4	19.3
	5%	DB	24.6	24.7	23.9	22.3	20.1	17.5	17.0	18.2	20.7	22.6	22.6	23.7
		MCWB	20.8	21.1	20.0	17.6	15.2	12.8	11.4	12.1	13.5	15.3	18.1	19.5
	10%	DB	23.7	23.8	23.1	21.3	19.0	16.5	15.9	16.9	18.9	20.3	21.1	22.6
		MCWB	20.3	20.5	19.6	17.3	14.7	12.3	11.0	11.8	13.7	15.6	17.7	19.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.1	23.2	22.8	20.6	18.3	16.1	14.9	15.6	17.4	19.0	20.6	22.3
		MCDB	25.0	25.7	24.4	22.7	19.8	18.5	16.9	18.1	21.0	23.4	23.3	25.1
	2%	WB	22.2	22.4	21.8	19.6	17.3	14.7	13.7	14.5	16.4	18.1	19.7	21.2
		MCDB	24.2	24.6	23.6	21.9	19.4	16.9	16.1	17.1	19.3	21.1	22.3	23.4
	5%	WB	21.6	21.8	21.0	18.8	16.4	13.9	12.7	13.6	15.6	17.2	18.9	20.5
		MCDB	23.7	23.8	22.9	21.2	18.9	16.2	15.2	16.5	18.4	20.2	21.4	22.6
	10%	WB	21.0	21.2	20.3	18.0	15.4	13.2	11.9	12.8	14.7	16.5	18.2	19.7
		MCDB	23.1	23.2	22.4	20.4	18.3	15.6	14.6	16.0	17.9	19.6	20.5	21.8
Mean Daily Temperature Range	5% DB	MDBR	5.7	5.6	5.9	6.4	6.4	6.0	6.5	7.1	7.3	7.3	6.4	6.0
		MCDBR	7.2	6.9	7.2	8.0	7.6	7.0	7.7	8.6	10.7	12.1	9.5	8.4
		MCWBR	3.9	3.9	4.5	5.0	4.8	4.5	4.7	4.9	5.1	5.2	4.6	4.4
	5% WB	MCDBR	6.0	6.2	5.9	6.8	6.4	5.7	6.3	7.2	7.9	8.0	7.2	6.6
		MCWBR	3.7	3.9	4.1	4.8	4.6	4.3	4.6	5.1	5.0	4.8	4.4	4.1
Clear-Sky Solar Irradiance	taub		0.408	0.381	0.354	0.327	0.311	0.299	0.292	0.302	0.333	0.351	0.380	0.382
	taud		2.400	2.512	2.569	2.619	2.611	2.641	2.654	2.618	2.525	2.505	2.455	2.469
	Ebn at Noon		928	933	922	892	853	839	869	908	928	952	949	958
	Edh at Noon		126	108	95	80	71	64	67	78	97	108	118	118
All-Sky Solar Radiation	RadAvg		6.32	5.43	4.67	3.71	2.94	2.28	2.72	3.57	4.81	5.65	5.95	6.46
	RadStd		0.52	0.45	0.34	0.31	0.16	0.19	0.22	0.20	0.30	0.42	0.48	0.45

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	-1	-82	+172	+80

Nomenclature: See separate page